

Deploying a Dockerized Node App for Customer Portals



Estimated time needed: 30 minutes

In this lab, you will set up and deploy a Node.js application coupled with MongoDB on Docker. Your objective is to replicate the customer portal repository utilized in previous exercises and commence the deployment of the customer portal application using Docker.

Learning Objective:

After completing this lab, you will be able to:

- Clone a Customer portal Node.js application from a GitHub repository
- Complete configuration of Dockerfile and docker-compose.yml
- Deploy the Node.js application on Docker

About Skills Network Cloud IDE

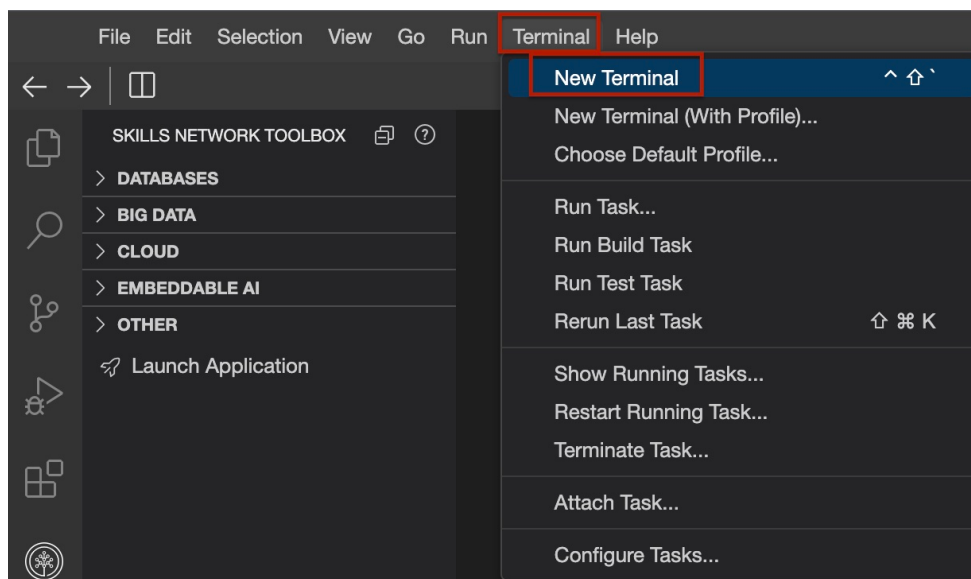
Skills Network Cloud IDE (based on Theia and Docker) provides an environment for hands on labs for course and project related labs. Theia is an open source IDE (Integrated Development Environment).

Important Notice about this lab environment

Please be aware that sessions for this lab environment are not persisted. Every time you connect to this lab, a new environment is created for you. Any data you may have saved in the earlier session would get lost. Plan to complete these labs in a single session, to avoid losing your data.

Task 1: Clone a Customer portal Node.js application from a GitHub repository

1. Open a terminal window by using the menu in the editor: Terminal > New Terminal.



2. Change to your project folder; if you are not in the project folder already.

```
cd /home/project
```

3. In the terminal, clone the git repository which has the starter code.

```
git clone https://github.com/ibm-developer-skills-network/abltc-backend-nodejs-customerportal.git
```

4. Change to the new directory that you just cloned.

```
cd abltc-backend-nodejs-customerportal
```

5. In the terminal, run the following command to install the node packages required.

```
npm install
```

Task 2: Configuration of Dockerfile and docker-compose.yml

1. Create an Dockerfile file using the command below.

```
touch Dockerfile
```

2. Paste the below code into Dockerfile.

```
FROM node:18.12.1-bullseye-slim
# Set the working directory in the container
WORKDIR /app
# Copy package.json and package-lock.json to the working directory
COPY package*.json ./
# Install app dependencies
RUN npm install
# Copy necessary files
COPY . .
# Expose port 3000 (adjust the port based on your application)
EXPOSE 3000
# CMD to run the application
CMD [ "node", "customer_app.js" ]
```

3. Create an docker-compose.yml file using below command.

```
touch docker-compose.yml
```

- Paste the below code into docker-compose.yml

```
version: '3'
services:
  mongodb:
    image: mongo
    ports:
      - "27017:27017"
    networks:
      - app-network

  app:
    image: customerapp
    ports:
      - "3000:3000"
    depends_on:
      - mongodb
    networks:
      - app-network

networks:
  app-network:
    driver: bridge
```

Task 3: Deploy the Node.js application on Docker

- Run the following command to build the Docker app

```
docker build . -t customerapp
```

```
theia@theiadocker-pallavir:/home/project/SocialMediaDashboard$ docker build . -t socialapp
[+] Building 18.5s (8/10)                                docker:default
=> => sha256:79bf84739156657c85440e6a55a3d77a7cac668f9c4c3c44005bc29bdc529db 1.21kB / 1.21kB 0.0s
=> => sha256:0c3ea57b6c560f83120801e222691d9bd187c605605185810752a19225b5e4d 1.37kB / 1.37kB 0.0s
=> => sha256:a26e85f2ac4163ff881438833dd8f4f6f7282b6f03c4fc651345f80305c4945 6.80kB / 6.80kB 0.0s
=> => sha256:3f4ca61aafcd4fc07267a105067db35c0f0ac630e1970f3cd0c7bf552780e 31.40MB / 31.40MB 4.8s
=> => sha256:128624e9c76ab0636e60e3fe2463e9a710346c68f47b4711912ff1df3c1a64b3 451B / 451B 1.4s
=> => extracting sha256:3f4ca61aafcd4fc07267a105067db35c0f0ac630e1970f3cd0c7bf552780e985 2.1s
=> => extracting sha256:00fde01815c92cc90586fcf531723ab210577a0f1cb1600f08d9f8e12c18f108 0.0s
=> => extracting sha256:723367e6ef2d175cd363a4ca96a65ac6fa3f388506de7c0489318e8d8ed59dd7 2.9s
=> => extracting sha256:907e6809ddf0f1b24bb5a93c1f0f01034e905fa3a4c47bdb47d7e90018472cef 0.1s
=> => extracting sha256:128624e9c76ab0636e60e3fe2463e9a710346c68f47b4711912ff1df3c1a64b3 0.0s
=> [2/5] WORKDIR /app 0.0s
=> [3/5] COPY package*.json ./ 0.1s
```

- The docker-compose.yml has been created to run two containers, one for Mongo and the other for the Node app. Run the following command to run the server:

```
docker-compose up
```

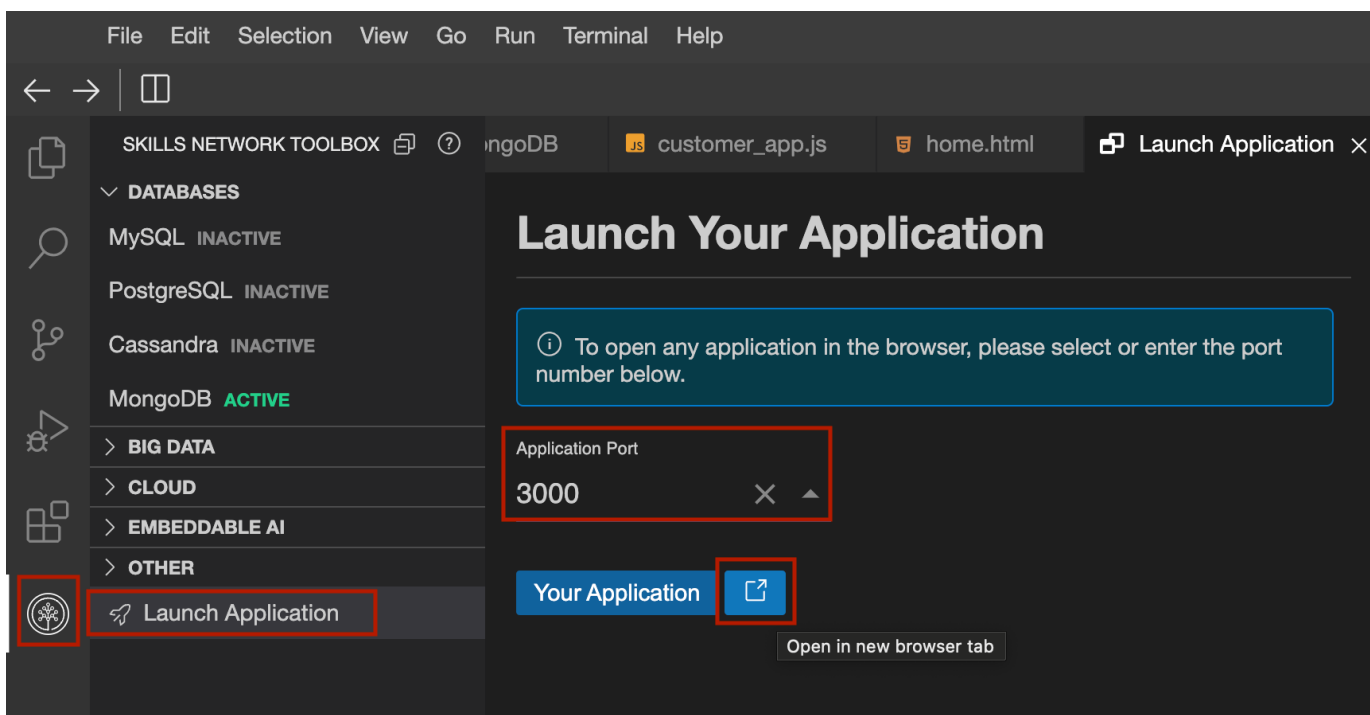
```
theia@theiadocker-pallavi: /home/project/abltc-backend-nodejs-customerportal$ docker-compose up
[+] Running 8/10
! mongodb 9 layers [██████████] 39.79MB/224.4MB Pulling
✓ a48641193673 Download complete
✓ 9cf348a38ee3 Download complete
✓ 3aa8507b7d76 Download complete
✓ 841ef8d499cd Download complete
✓ 97788ce9e1a0 Download complete
✓ 194a4afa4855 Download complete
✓ becfac27bfec Download complete
! 63d2c40642ff Downloading 39.79MB/224.4MB
✓ 1634c5425c04 Download complete
```

3. Click on the button below to launch the application.

Customer Portal

▼ Click here if the button above doesn't work

1. Click on the Skills Network icon in the left tool bar.
2. Choose Launch Application.
3. Enter the port number 3000 on which the application is running.
4. Click on the icon, as shown in the image, to open the customer portal in a new tab.



Authors

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